TradeSentinel

Should we fund a pilot of AI-assisted support triage for trading issues?

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Recommendation: pilot in shadow mode only, and do not scale yet

Why go ahead

- The decision repeats on every ticket and its cost is measurable
- AI read reports far better than rules: 90% against 45.7%
- With controls, 97.6% of critical issues were escalated

Why not more than a pilot

- Three critical issues were under-triaged in testing
- Time saved has not been measured
- All data so far is synthetic

The ask

- About USD 60,000 build (assumed)
- Fix and re-test, then 8 weeks in silent shadow
- One flow, one client tier, one venue
- Stop conditions agreed in advance

A missed execution fault costs far more than a needless escalation

- Reports arrive as free text: “my orders are stuck”
- Each needs reading, sizing and routing: 10 to 20 minutes (assumed)
- The queue grows fastest in volatile markets
- One fault gets opened many times and worked separately
- Quality depends on who is on shift

USD per incident (assumed)	Escalate	Do not escalate
Genuinely critical	+5,000	−50,000
Actually minor	−150	+40

About 0.3% chance of being critical is enough to justify escalating. Even with far less extreme inputs, break-even stays under 3.5%.

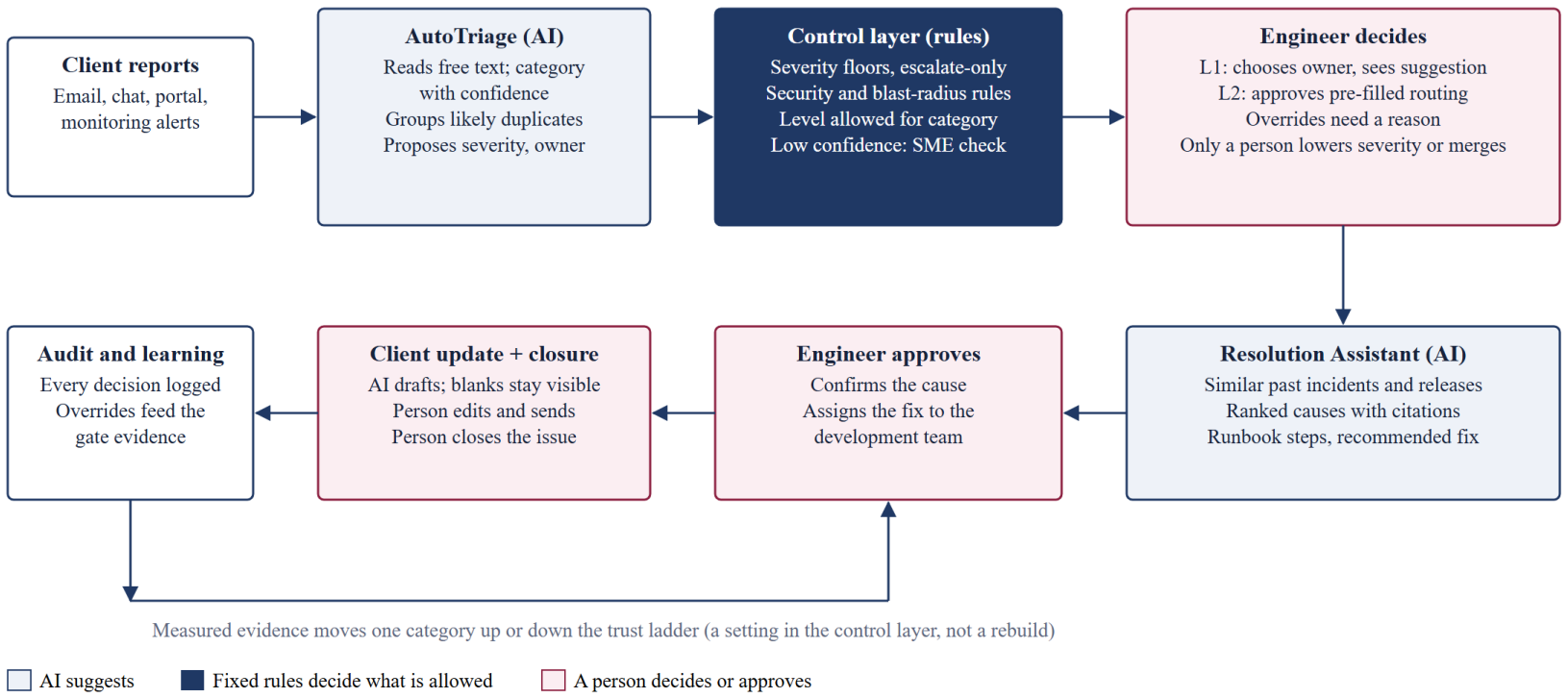
The user is a support engineer deciding under pressure, and rules alone cannot read client language

Job to be done

When I am on shift and reports arrive in clients' own words during a volatile market, I want to know quickly which are the same fault, how serious each is and what probably caused it, so that I can protect clients without chasing every report by hand.

Alternative	Why it falls short
Keyword rules	Break on reworded and understated reports
Dashboards	Show machine signals, cannot read clients
More engineers	Adds cost, not consistency
AI model alone	Missed 7 of 41 critical issues in our test
AI + control layer	**Missed 1 of 41; a real gap remains**

AI reads and matches; rules set the limits; a person makes the decisions that matter



Trust ladder

- L0 Shadow: silent, recorded
- L1 Suggest: person decides
- L2 Assisted: one-click approval
- L3 Limited auto: not earned yet

Never allowed

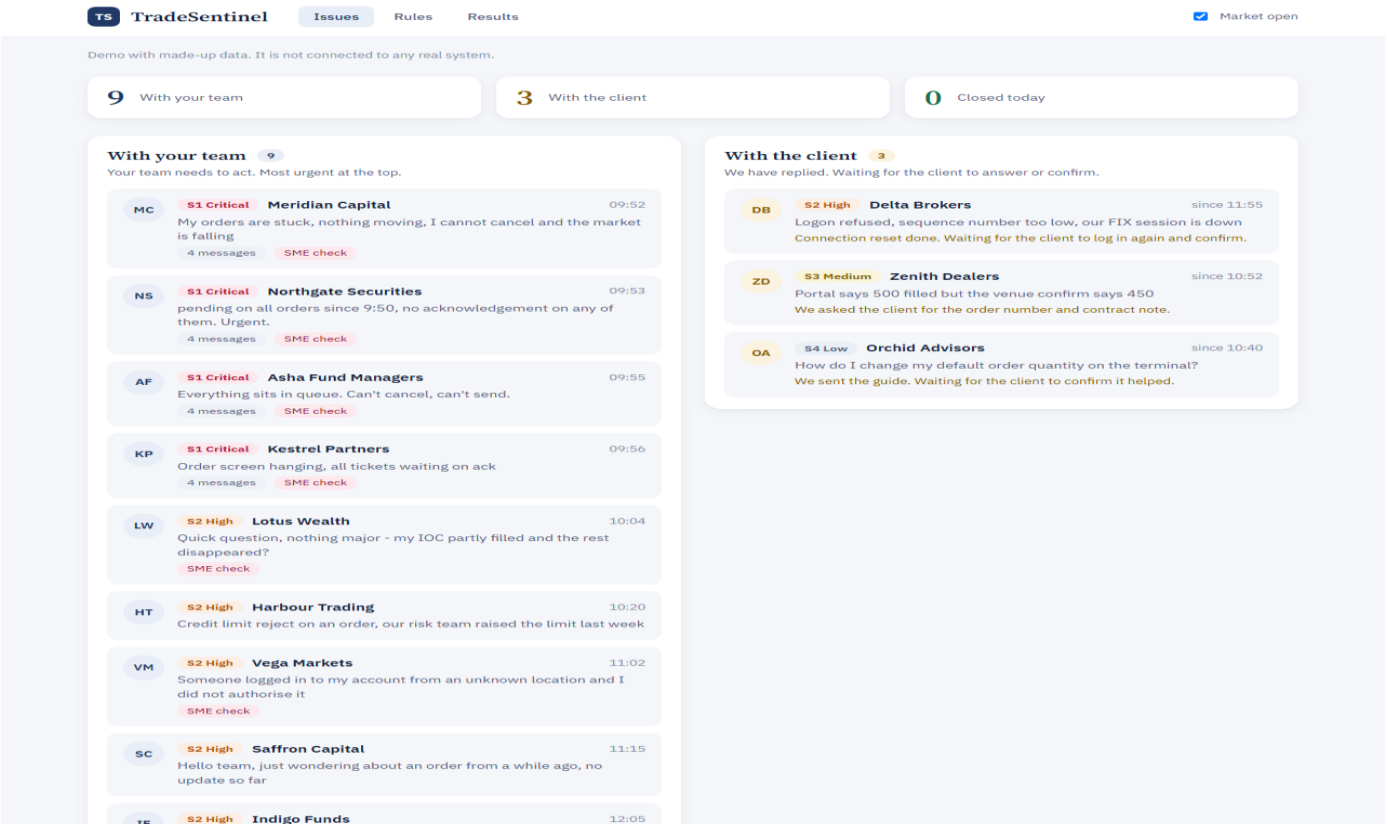
Lowering severity, closing, merging, production changes, or sending unapproved or uncited updates.

MVP: order-entry and execution issues, one client tier, one venue.

We made hard choices on scope, sequence and controls

Decision	What we chose	What we gave up	Why
First wedge	Riskiest flow: order and execution	An easier early win	If controls hold here, they likely hold elsewhere
Build vs buy	API for models; build the control layer	Model ownership	Learning speed now; our SLAs and audit are specific to us
Held at L1	Execution failures, on purpose	Time saved on that category	It has the highest cost of error
Rejected	Autonomous fixes; client chatbot	A flashier demo	One bad action could outweigh a year of savings
Prioritization	Controls before features; RICE for the rest	Faster feature delivery	Controls are a precondition for launch

The prototype shows who must act next, and works end to end



[Live demo: jodhdeep10.github.io/tradesentinel-console](jodhdeep10.github.io/tradesentinel-console)

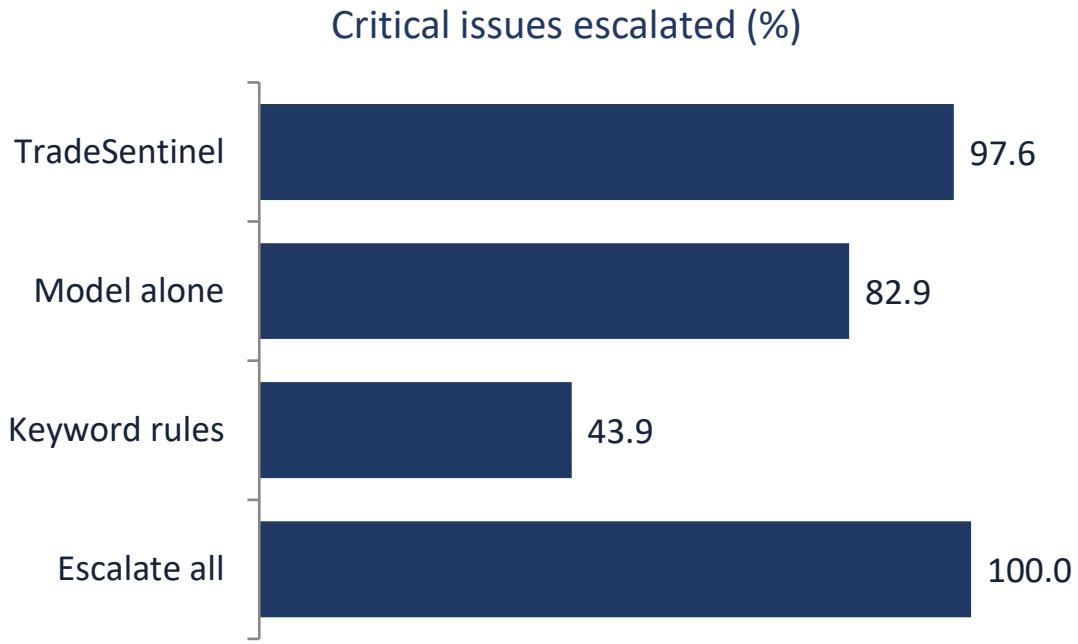
What the demo proves

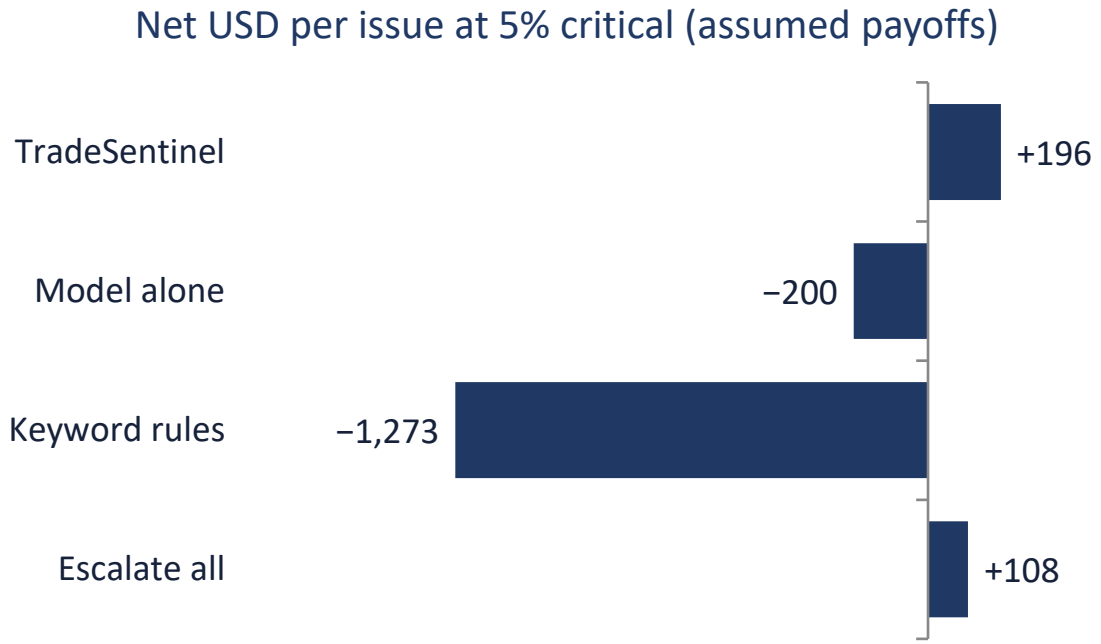
- Two lanes: with your team, with the client
- Doubt stated plainly; reasons one click away
- Autonomy changes by configuration
- Reply is editable; blanks lock Send

What it does not prove

- Language model quality: a local classifier stands in
- Results on real tickets
- Time saved or engineer trust

The control layer is what makes this worth funding





90% category accuracy (rules 45.7%)

96.8% true cause in top three (target 70%)

5 of 5 guardrail checks held on every ticket

Synthetic data; one author wrote the data and the tests, so results are likely optimistic. 95% interval on critical escalation: 87.4% to 99.6%.

It failed where we most feared: “Not urgent, but…”

“Not urgent, but my limit order that should have traded at the touch shows nothing done.”

True severity S2. System said S3. Wrong category at 82% confidence, so no control fired.

Gate 1 measure	Target	Result	Status
Correct cause in top hypotheses	At least 70%	96.8%	Met (likely flattered)
Critical issues under-triaged	Near zero	3 of 41 (1 below escalation)	Not met
Time to first correct triage	Under 3 minutes	Not measured	Needs real users

Next: add a rule for orders with no visible outcome, cap confidence on thin reports, and re-test on a new set written by someone else.

Inference is cheap; false escalations are not, so the case rests on avoided misses

Per issue, 1,000 issues a month	Conservative	Base	Optimistic
Time saved	USD 7	USD 30	USD 40
Inference	−USD 0.10	−USD 0.05	−USD 0.05
Extra false escalations	−USD 19.67	−USD 12.54	−USD 5.42
Run and maintain	−USD 7.00	−USD 7.00	−USD 7.00
Net per issue	**−USD 19.77**	**USD 10.41**	**USD 27.54**
Payback on build	None from time alone	5.8 months	2.2 months

1 miss

avoided about every 2.8 months covers the conservative shortfall of USD 19,765 a month

Break-even at about 403 issues a month in the base case. A very small desk may not need this.

Funded internally for the pilot; a SaaS offer to other brokers only if reuse is proven. All amounts are illustrative assumptions.

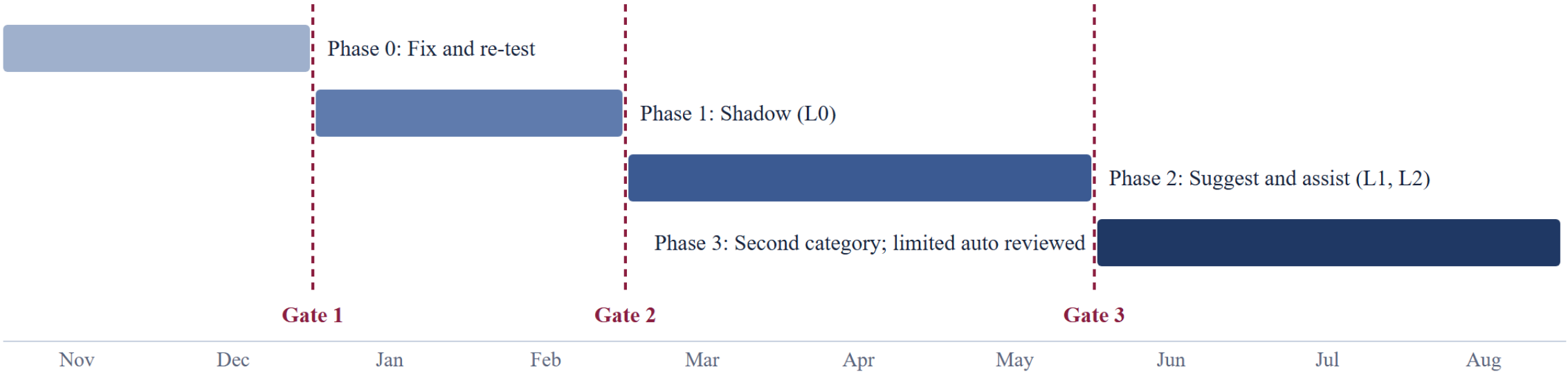
Specific risks have specific controls, and six things block launch

Risk	How it shows up here	Control
Bias	Polite or non-native wording gets lower severity	Fairness test by style and client size before Gate 2
Privacy	Tickets hold client names, orders and positions	Masking; private route; no-training contract
Hallucination	An invented cause reaches clients during an incident	Citations; visible blanks; Send lock; approval
Misuse	“Urgent” gaming; prompt injection in ticket text	Floors ignore such words; rules sit outside the model
Rubber-stamping	Tired engineers approve without reading	L1 not pre-filled; review time tracked; samples

Launch blockers

- Critical miss in shadow with no tested fix
- Client data cannot be protected
- Any uncited send possible
- Material fairness gap
- Review time collapses, errors do not
- Operations or risk declines

Each phase must earn the next one through a gate



Dates are illustrative. A phase starts only when the gate before it is passed, so a failed gate moves everything to its right.

Gate 1

New test set with no critical issue below the escalation line; timed trial under 3 minutes

Gate 2

Real tickets match experts; no S1 called below S2; fairness and masking verified

Gate 3

Positive net payoff; ECE below 0.05; operations and risk sign-off

Decision needed today: approve the shadow pilot

We recommend

- Pilot in shadow mode only; do not scale
- One flow, one client tier, one venue
- About USD 60,000 build (assumed)
- Phase 0 fixes the understated-report gap
- 8 weeks silent on masked real tickets
- Gate 2 decides: L1, stay in shadow, or stop

We would stop if

- Critical issues still fall below the escalation line in shadow
- Real-ticket accuracy does not clearly beat rules
- Client data cannot be protected
- A useful but unsafe system stays an assist feature